

Geography—Rivers of India and the World

Subject: Geography

A river is a natural channel through which water flows from a higher area toward a lower area, usually ending in a lake, another river, or the sea. The beginning of a river is called its source. A source may be a spring, glacier, lake, or wet highland. Small streams join to form tributaries, and tributaries join the main river. The land drained by a river and its tributaries is called a drainage basin or watershed. A river's mouth is the place where it enters a sea, ocean, lake, or another river.

The work of a river changes from its upper course to its lower course. Near its source, the land is often steep, so the river has high vertical erosion and cuts downward into the landscape. Rapids, gorges, V-shaped valleys, and waterfalls may form. A waterfall develops when harder rock lies above softer rock, when a fault creates a sudden step, or when a river crosses a steep change in elevation. The softer material is worn away faster, leaving a ledge. Water plunges over the ledge, and a plunge pool may form below it. Over time, the hard rock can break and the waterfall may move upstream, creating a gorge.

In the middle course, the slope is usually gentler and the river carries more water. Erosion on the outside of bends and deposition on the inside create meanders. A meander may become so curved that the river cuts through the narrow neck, leaving an oxbow lake. In the lower course, the river is wider and deeper, and deposition becomes important. Floodplains, levees, braided channels, and deltas can develop. A delta forms when a river slows at its mouth and deposits sediment faster than waves and tides can remove it. River floods can damage homes and crops, but floodplain sediments can also make farmland fertile.

India has several important river systems. The Ganga begins in the Himalayas and flows across the northern plains before reaching the Bay of Bengal through a large delta shared with the Brahmaputra system. The Yamuna is a major tributary of the Ganga. The Brahmaputra rises in the Tibetan region, flows through Arunachal Pradesh and Assam, and joins the Ganga system. The Indus and its tributaries are important in the northwestern region. Peninsular rivers include the Godavari, Krishna, Kaveri, Mahanadi, Narmada, and Tapi. Many Himalayan rivers are perennial because they receive water from snow, glaciers, and rainfall, while several peninsular rivers depend more strongly on seasonal monsoon rainfall.

Around the world, famous rivers include the Amazon, Nile, Yangtze, Mississippi–Missouri, Yenisei, Yellow River, Congo, Mekong, Danube, and Volga. The Amazon carries an exceptionally large volume of water and flows through a vast tropical basin. The Nile is associated with the development of ancient Egyptian civilization. The Yangtze is the longest river in Asia, while the Mississippi–Missouri system drains a large part of North America. The Congo flows through central Africa and is notable for its deep channel and large discharge. Rivers differ greatly: some are long, some carry more water, some are very wide, and some are short

mountain streams. River size must therefore be described using more than one measurement, such as length, drainage area, width, depth, or discharge.

Rivers support drinking water, irrigation, transport, hydroelectric power, fishing, wildlife, and cultural life. Dams and reservoirs can provide electricity and regulate water, but they may disrupt fish migration, trap sediment, displace communities, and change downstream ecosystems.

Pollution from sewage, chemicals, plastics, and farm runoff can reduce water quality.

Sustainable river management protects wetlands, treats wastewater, maintains environmental flow, restores vegetation along banks, and reduces construction in dangerous flood zones.

Understanding how rivers erode, transport, and deposit material helps people use river landscapes while reducing risk.